File 60: The Coldwater Spawning Sanctuary Masterclass & Advanced Geomorphic Trout Habitat Restoration Standards

> ***Trout Ecology and Engineering Standard***: *This document establishes the scientific, geomorphic, and electrical engineering standards for achieving the highest ecological level of coldwater stream restoration. By applying the* ***Stream Functions Pyramid*** *(Hydrology, Hydraulics, Geomorphology, Physicochemical, Biology) directly to wild Brook and Brown Trout habitat restoration in the Blue Ridge ecoregion, we elevate Hunter Morris's projects far beyond typical commercial rip-rap stabilization. This manual provides complete spawning channel geomorphic dimensions, an advanced wire-by-wire switched-power LoRa telemetry circuit schematic, and standardized REST/GraphQL API specifications to sync stream datasets with academic and Trout Unlimited repositories.*

I. The Stream Functions Pyramid for Trout Restoration

To ensure our restoration projects satisfy the highest standards of conservation science, we evaluate and design all fluvial channels using the **Stream Functions Pyramid** (Harman et al., 2012). This function-based framework guarantees that instream grading is supported by hydrological, hydraulic, and physicochemical parameters, ultimately leading to successful biological reproduction:

```
                 [THE STREAM FUNCTIONS PYRAMID FOR WILD TROUT]

                            /=======================\
                           /    Level 5: BIOLOGY     \  <-- Redds (spawning), macroinvert
ebrate
                          / (Brook & Brown Spawning)  \     biomass, EPT colonization rat
es.
                         /=============================\
                        /   Level 4: PHYSICOCHEMICAL    \  <-- Stream Temp < 65°F, Dissol
ved Oxygen
                       / (Thermal Shading & DO Regimes)  \    > 7.0 mg/L in gravel pore s
paces.
                      /===================================\
                     /      Level 3: GEOMORPHOLOGY         \  <-- Pool-riffle sequencing,
 bankfull stable
                    /  (Rosgen Classification & Gravels)    \     3:1 regrading, Rhododen
dron maximum geogrid.
                   /=========================================\
                  /           Level 2: HYDRAULICS             \  <-- Flow velocity 0.15–0
.45 m/s, pool shear
                 /     (1D/2D HEC-RAS Shear Stress Models)     \     stresses, flood-plai
n floodplain connectivity.
                /===============================================\
               /              Level 1: HYDROLOGY                 \  <-- LoRa mesh stage l
ogging, watershed drainage,
              /       (Rainfall-Runoff & Baseflow Logging)        \     HUC-8 basin geomo
rphic discharge curves.
```

```
               /=========================================================\
```

The Five Functional Levels Applied:

1. **Level 1: Hydrology (Watershed Flow Dynamics)**
- *Metric*: Continuous stage logging via hydrostatic transmitters feeding base gateways.
- *Trout Context*: Establishes baseline baseflow discharge curves, ensuring headwater streams maintain dry-season flow volumes required for egg survival in spawning redds.
2. **Level 2: Hydraulics (Local Fluid Mechanics)**
- *Metric*: Bankfull shear stress (*tau*), flow velocity (*v*), and floodplain connectivity.
- *Trout Context*: Riffle flow velocities must remain in the $0.15\text{ m/s} **-** 0.45\text{ m/s} *sweep range. High-gradient pool velocities are regulated to provide resting micro-habitats, keeping shear stress below gravel movement thresholds ($1.46\text{ N/m}^2 < \tau < 21.85\text{ N/m}^2*) to prevent redd washout.
3. **Level 3: Geomorphology (Physical Channel Form)**
- *Metric*: Rosgen Classification (Type A-B channels), pebble-count distribution (*D_{50}* median grain size), and lateral bank stability.
- *Trout Context*: Regrading vertical failing banks to stable 3:1 slopes, reinforced with geomechanical *Rhododendron maximum* root-wads. Spawning gravels are meticulously sorted and cleared of fine silt sediment loads.
4. **Level 4: Physicochemical (Water Quality & Chemistry)**
- *Metric*: Continuous water temperature loggers, optical dissolved oxygen (DO), and NTU turbidity.
- *Trout Context*: Maintains summer thermal water regimes below $65^\circ\text{F} *($18.3^\circ\text{C}*) via dense canopy cover, keeping dissolved oxygen levels above 7.0 mg/L directly in the spawning gravel substrate.
5. **Level 5: Biology (Ecology & Reproduction)**
- *Metric*: Redds count, EPT macroinvertebrate colonization indices (Mayfly, Stonefly, Caddisfly), and trout biomass.
- *Trout Context*: The ultimate metric of success. Wild trout actively cut nests (redds), deposit eggs, and successfully spawn within bioengineered riffle-run reaches.

II. Spawning Sanctuary Design Standards

To build the highest-tier spawning sanctuaries, Hunter Morris's restoration crew utilizes these precise geomorphic and ecological design standards:

1. Spawning Riffle Geomorphic Parameters

- **Media Gravel Sorting**: Spawning gravels must be native, washed, and sorted to match species-specific spawning parameters:
- *Brook Trout*: *D_{50}* median grain size of **12mm to 25mm** (coarse pea gravels to small pebbles).
- *Brown Trout*: *D_{50}* median grain size of **15mm to 40mm** (medium pebbles).
- **Substrate Depth**: Gravel beds must have an active depth of **10cm to 30cm** over a firm, stable base.

- **Flow Hydraulics (Redd Velocity)**:
- *Water Depth*: Riffle spawning zones are designed with depths of **10cm to 45cm** under normal base flows.
- *Velocity*: Stream velocities directly above the gravel bed must remain between **0.15 m/s and 0.45 m/s**.
- **Subsurface Interstitial Flow**: Spawning beds are built immediately upstream of riffle crests (where downwelling occurs), pushing oxygen-rich water ($>7.0\text{ mg/L}$ DO) through the gravel pore spaces to nourish incubating eggs.

2. Thermal Shading and Canopy Density Design

Direct solar radiation raises stream temperatures, driving wild trout out of shallow creeks. To ensure stream temperatures remain cool, we design a multi-tiered **Riparian Shade Canopy** using the **Solar Shading Shading Equation**:

$$S_{\text{attenuation}} = 1 - e^{-k \cdot \text{LAI}}$$

- *Where*: k is the canopy extinction coefficient (typically **0.55** for broadleaf/Rhododendron maximum canopy), and LAI is the Leaf Area Index.
- *Design Target*: To achieve a **96% solar radiation reduction** ($S_{\text{attenuation}} = 0.96$), we plant a dense broadleaf canopy targeting a minimum **LAI of 5.8**.
- *Planting Matrix*: We transplant mature, field-sourced *Rhododendron maximum* and *Alnus serrulata* (Tag Alder) at a spacing density of **1.5 meters on-center** directly along the stream bank edge.

3. Pool-to-Riffle Sequencing Ratios

- High-gradient Blue Ridge stream reaches (Rosgen B3-B4 types) require a pool-to-riffle spacing ratio of **5 to 7 bankfull widths**.
- Pools are excavated immediately downstream of geomechanical rock J-hook vanes to provide deep, thermal refuges (minimum depth **1.2m to 1.8m**) with slow resting currents ($<0.1\text{ m/s}$).

III. Switched-Power Hardware Schematics & Component Interface Wiring

To ensure long-term, off-grid telemetry operation on a small **3.7V 3200mAh LiFePO4 battery**, we cannot leave our high-draw 12V Modbus sensors powered continuously. We utilize a **Switched-Power MOSFET Rail** controlled by a microcontroller GPIO pin to apply 12V boost power only during measurement windows.

1. Complete wire-by-wire Schematic

This ASCII circuit diagram shows exactly how to connect the WisBlock/ESP32, MAX485 transceiver, MT3608 boost regulator, and AO3401 P-Channel MOSFET switch:

```
                                [STREAM NODE SWITCHED-POWER SCHEMATIC]

  3.7V battery (LiFePO4)
      +     -
     [+]---[-]--------------------------------------------------------------------+
      |     |                                                                     |
      |     v (GND)                                                               v
      |   +----------------------------------------+                       +-------
----+
      +-->| IN+    [MT3608 12V Boost Regulator]   |                       |  MAX48
5   |
          | IN-                                    |                       |
```

```
    |
          | OUT+ (Boosted 12V Rail)   OUT- (GND)  |                               |  VCC (
3.3)|<--+ 3.3V (WisBlock)
          +--|--------------------------|---------+                               |  GND
    |<--+ GND (WisBlock)
             |                          |                                         |  RO (R
X)  |--> WisBlock GPIO 16
             v                          v (GND)                                   |  DI (T
X)  |<-- WisBlock GPIO 17
      +------------+                    |                                         |  DE /
RE  |<-- WisBlock GPIO 4 (Shorted)
      | [AO3401]   |                    |                                         |  A (+)
    |--+ RS485 Differential A+
      | P-MOSFET   |                    |                                         |  B (-)
    |--+ RS485 Differential B-
      | Source     |                    |                                         +-------
----+
      | Drain      |--+ 12V Switched    |
      | Gate       |  |   VCC to Sensors|
      +----|-------+  |                 |
           |          v                 v
           |     +----------+      +----------+
           |     | DS-DO    |      | TJ-F04   |  (12V Modbus Sensors)
           |     | Optical  |      | Pressure |
           |     | DO       |      | Level    |
           |     +----------+      +----------+
           |
           +---------------------+
                                 |
                                 v
                              [2N2222 NPN BJT Pre-Driver]
                              Collector
                              Emitter   ======> GND
                              Base      <====== WisBlock GPIO 12 (via 1k Ohm Resistor)
```

2. Component Pin-to-Pin Connection Schedule

Use this exact wiring layout to assemble the streamside PCB node:

Source Component	Pin Name	Target Component	Pin Name	Connection Wire Purpose
WisBlock / ESP32	**3.3V OUT**	**MAX485 Chip**	**VCC**	Provides 3.3V logic power to transceiver.
WisBlock / ESP32	**GND**	**MAX485 Chip**	**GND**	Common ground line.
WisBlock / ESP32	**GPIO 16**	**MAX485 Chip**	**RO (Receive Out)**	Telemetry Serial RX line.
WisBlock / ESP32	**GPIO 17**	**MAX485 Chip**	**DI (Driver In)**	Telemetry Serial TX line.
WisBlock / ESP32	**GPIO 4**	**MAX485 Chip**	**DE & RE (Shorted)**	half-duplex Transmit/Receive enable pin.

Source Component	Pin Name	Target Component	Pin Name	Connection Wire Purpose
WisBlock / ESP32	**GPIO 12**	**2N2222 BJT**	**Base (via 1k resistor)**	Power switch trigger pin.
MT3608 Boost	**OUT+ (12V)**	**AO3401 P-MOS**	**Source**	Boosted 12V power supply feed.
AO3401 P-MOS	**Gate**	**2N2222 BJT**	**Collector**	Pulled LOW to activate P-MOS power rail.
AO3401 P-MOS	**Drain**	**All Sensors**	**VCC (Industrial 12V)**	Switched 12V power output to sensors.
2N2222 BJT	**Emitter**	**System GND**	**GND**	Ground for BJT switch.
MAX485 Chip	**A (+)**	**All Sensors**	**Yellow/Green (A+)**	RS485 Differential A+ line.
MAX485 Chip	**B (-)**	**All Sensors**	**Blue/White (B-)**	RS485 Differential B- line.

IV. Trout Telemetry REST & GraphQL Integration APIs

Our streamside base gateway operates a private REST API database replication layer that synchronizes environmental data with national conservation repositories.

1. Unified Biological Correlation REST Ingress

Endpoint: `POST https://science.tu.org/api/v1/ingress/blue_ridge/biological-sync`
This API payload maps real-time water quality parameters directly against monthly macroeconomic macroinvertebrate colonization counts, validating geomorphic restabilization in real time:


```
{
  "ingress_token": "brsr_science_token_secure_2026",
  "metadata": {
    "station_id": "BRSR-LO-101",
    "station_moniker": "Upper Toccoa headwater Spawning Sanctuary 2",
    "huc_8_watershed": "06020003",
    "stream_name": "Anderson Creek",
    "county": "Fannin County",
    "state": "GA"
  },
  "water_quality_averages": {
    "sampling_window_start": "2026-05-01T00:00:00Z",
    "sampling_window_end": "2026-05-31T00:00:00Z",
    "mean_temperature_f": 57.42,
    "max_temperature_f": 62.15,
    "mean_dissolved_oxygen_mg_l": 9.45,
```

```
    "mean_turbidity_ntu": 12.8,
    "sediment_status": "EXCELLENT_GRAVEL_CLARITY"
  },
  "biological_field_metrics": {
    "benthic_sample_date": "2026-05-28T14:30:00Z",
    "macroinvertebrate_ept_index": 22,
    "stoneflies_count_pct": 28.5,
    "mayflies_count_pct": 35.0,
    "caddisflies_count_pct": 20.5,
    "taxa_richness": 18,
    "redd_nest_count": 8,
    "brook_trout_spawning_verified": true
  },
  "geomorphic_validation": {
    "bankfull_width_m": 4.12,
    "mean_bank_stability_rating": "STABLE_BIOENGINEERED_RHODODENDRON",
    "riffle_shear_stress_n_m2": 14.85,
    "pebble_count_d50_mm": 18.2
  }
}
```


2. UGA Warnell GraphQL Telemetry Sourcing API

Warnell researchers pull localized ecoregion data using a high-performance **GraphQL API** to build climate water thermal warning models.

Warnell GraphQL Query Structure:

```
query GetTroutThermalRegime($stationId: String!, $huc8: String!, $startDate: DateTime!) {
  streamMonitoringStation(stationId: $stationId, huc8: $huc8) {
    stationName
    elevationMeters
    geomorphicReachType
    hourlyLogs(from: $startDate) {
      timestamp
      waterQuality {
        temperatureC
        dissolvedOxygenMgL
        oxygenSaturationPct
        turbidityNtu
      }
      hydrology {
        stageLevelMeters
        dischargeCfs
      }
      hootOwlAlertActive
    }
  }
}
```

Warnell GraphQL JSON Response:


```
{
  "data": {
    "streamMonitoringStation": {
      "stationName": "Anderson Creek Bio-Reserve 2",
      "elevationMeters": 680.5,
      "geomorphicReachType": "Rosgen B3 Spawning Riffle",
      "hourlyLogs": [
```

```
        {
          "timestamp": "2026-05-30T18:00:00Z",
          "waterQuality": {
            "temperatureC": 14.85,
            "dissolvedOxygenMgL": 9.42,
            "oxygenSaturationPct": 91.2,
            "turbidityNtu": 10.5
          },
          "hydrology": {
            "stageLevelMeters": 0.412,
            "dischargeCfs": 18.5
          },
          "hootOwlAlertActive": false
        }
      ]
    }
  }
}
```


Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.